

Ex Relations on sets:

Let A and B be two sets as follows:

$$A = \{\text{Calcutta, Patna, Lucknow, Chennai}\}$$

$$\text{and } B = \{\text{West Bengal, Bihar, Uttar Pradesh, Tamil Nadu}\}$$

There is a relation 'is a capital of' between the elements of the sets A and B . If R is used for the relation 'is a capital of', then the above information can be written as Calcutta R West Bengal, Patna R Bihar, etc.

The above information can also be written as a set of ordered pairs R where

$$R = \{(\text{Calcutta, West Bengal}), (\text{Patna, Bihar}), (\text{Lucknow, Uttar Pradesh}), (\text{Chennai, Tamil Nadu})\}$$

$$= \{(x, y) : x \in A, y \in B, x R y\}$$

Thus, the relation 'is a capital of' from a set A to B gives rise to a subset R of $A \times B$ such that $(x, y) \in R$ if and only if $x R y$.

Definition: Let A and B be two sets. A relation from A to B is a subset of the Cartesian product $A \times B$. Suppose R is a relation from A to B . Then R is a set of ordered pairs (a, b) where $a \in A$ and $b \in B$. Every such ordered pair is written as $a R b$ and read as 'a is related to b by R'. If $(a, b) \notin R$, then a is not related to b by R and is written as $a \not R b$.

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Note: R is called a binary relation from A to B since the elements of the set R are ordered pairs.

Ex:

If $A = \{1, 2, 5\}$ and $B = \{2, 4\}$, then

$$A \times B = \{(1, 2), (1, 4), (2, 2), (2, 4), (5, 2), (5, 4)\}$$

Let R be ' $x < y$ '

Then, $R = \{(1, 2), (1, 4), (2, 4)\}$

Domain and Range:

The set $\{a \in A : (a, b) \in R \text{ for some } b \in B\}$ is called the domain of R and denoted by $\text{Dom}(R)$.

The set $\{b \in B : (a, b) \in R \text{ for some } a \in A\}$ is called the range of R and denoted by $\text{Ran}(R)$.

Thus, the domain of a relation R is the set of all the first element of the ordered pairs which belong to R and the range of R is the set of second elements.

Ex: Let $A = \{2, 3, 4\}$ and $B = \{3, 4, 5\}$. List the elements of each relation R defined below and the domain and range.

- a) $a \in A$ is related to $b \in B$, that is, aRb if and only if $a < b$
- b) $a \in A$ is related to $b \in B$, that is, aRb if a and b are both odd numbers.

→

a) $R = \{(2, 3), (2, 4), (2, 5), (3, 4), (3, 5), (4, 5)\}$

$$\text{Dom}(R) = \{2, 3, 4\}$$

$$\text{Ran}(R) = \{3, 4, 5\}$$

b) $R = \{(3, 3), (3, 5)\}$

$$\text{Dom}(R) = \{3\}$$

$$\text{Ran}(R) = \{3, 5\}$$

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Ex: Let $S = \{x, y\}$ and S^2 is the set of all words of length 2,

i) Find the elements of S^2 .

ii) The relation R on S^2 is defined by $v R w$ means that the first letter in v is the same as the first letter in w when v and w are in S^2 . Write R as a set of ordered pairs.

$$\rightarrow S^2 = \{xx, xy, yx, yy\}$$

$$R = \{(xx, xy), (xy, xx), (yx, yy), (yy, yx), (xx, xx), (yy, yy)\}$$

Total number of distinct relations from a set A to a set B :

Let the number of elements of A and B be m and n respectively. Then the number of elements of $A \times B$ is mn .

Therefore, the number of elements of the powerset of $A \times B$ is 2^{mn} . Thus, $A \times B$ has 2^{mn} different subsets.

Now, every subset of $A \times B$ is a relation from A to B .
Hence, the no. of different relations from A to B is 2^{mn} .

Some operations on sets:

Since, binary relations are sets of ordered pairs, all set operations can be done on relations. The resulting sets contain ordered pairs and are, therefore, relations. If R and S denote two relations, then

$$x(R \cap S)y = xRy \cap xSy$$

$$x(R \cup S)y = xRy \cup xSy$$

$$x(R - S)y = xRy \cap x \notin Sy$$

$$\text{and } x(R')y = x \notin Ry$$

Ex: If $A = \{x, y, z\}$, $B = \{x, y, z\}$, $C = \{x, y\}$ and $D = \{y, z\}$, R is a relation from A to B defined by $R = \{(x, x), (x, y), (y, z)\}$ and S is a relation from C to D defined by $S = \{(x, y), (y, z)\}$.
Find R' , $R \cup S$, $R \cap S$ and $R - S$.

$$\rightarrow A \times B = \{(x, x), (x, y), (x, z), (y, x), (y, y), (y, z), (z, x), (z, y), (z, z)\}$$

$$\text{Hence, } R' = \{(x, z), (y, x), (y, y), (z, x), (z, y), (z, z)\}$$

$$R \cup S = \{(x, x), (x, y), (y, z)\}$$

$$R \cap S = \{(x, y), (y, z)\}$$

$$R - S = \{(x, x)\}$$

Types of Relations in a Set:

Inverse Relation:

Let R be any relation from a set A to a set B . The inverse of R , denoted by R^{-1} is the relation from B to A which consists of those ordered pairs which, when reversed, belong to R ; that is,

$$R^{-1} = \{(b, a) : (a, b) \in R\}$$

$$\text{Consequently } x R y = y R^{-1} x$$

For ex., let A be the set of all living people.

We define relations B and C on A as follows:

$$B = \{(x, y) : x \text{ is a parent of } y\}$$

$$C = \{(y, x) : y \text{ is a child of } x\}$$

Then each of B and C is the inverse of other, written as $B = C^{-1}$ and $C = B^{-1}$.

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Similarly, on the set of real numbers, the relation $<$ is the inverse of the relation $>$.

Note:

There is a difference between R' and R^{-1} . The relation R' contains all elements not in R , R^{-1} contains all elements of R , except that their order is reversed.

For ex., if R is the relation $<$, then R^{-1} is the relation $>$, because $a < b$ if and only if $b > a$. On the other hand, R' is the relation $\geq$ because if $x < y$ is false then $x \geq y$.

Ex: Let $A = \{2, 3, 5\}$ and $B = \{6, 8, 10\}$ and define a binary relation R from A to B as follows:

For all $(x, y) \in A \times B$, $(x, y) \in R \Leftrightarrow x/y$ (x divides y).

Write each R and R^{-1} as a set of ordered pairs.

→ Here $2 \in A$ divides $6 \in B$, then $2 R 6$. Similarly, $2 R 8$, $2 R 10$, $3 R 6$, $5 R 10$

Therefore, $R = \{(2, 6), (2, 8), (2, 10), (3, 6), (5, 10)\}$

and $R^{-1} = \{(6, 2), (8, 2), (10, 2), (6, 3), (10, 5)\}$

Note:

(1) $\text{Dom}(R) = \text{Ran}(R^{-1}) = \{2, 3, 5\}$

$\text{Ran}(R) = \text{Dom}(R^{-1}) = \{6, 8, 10\}$

(2) If R is any relation, then $(R^{-1})^{-1} = R$.

Moreover, if R is a relation on A , then R^{-1} is also a relation on A .

Identity Relation:

A relation R in a set A is said to be identity relation, generally denoted by I_A , if

$$I_A = \{(x, x) : x \in A\}$$

Ex: Let $A = \{1, 2, 3\}$, then $I_A = \{(1, 1), (2, 2), (3, 3)\}$ is an identity relation in A .

Properties of Relations:

A relation R on a set A satisfies certain properties.

These properties are defined as follows:

Reflexive Relation:

- A relation R on a set A is reflexive if aRa for every $a \in A$, that is, if $(a, a) \in R$ for every $a \in A$. This simply means that each element a of A is related to itself.
- For ex.,
 - (a) If $R_1 = \{(1, 1), (1, 2), (2, 2), (2, 3), (3, 3)\}$ be a relation on $A = \{1, 2, 3\}$, then R_1 is reflexive relation since for every $a \in A$, $(a, a) \in R_1$.
 - (b) If $R_2 = \{(1, 1), (1, 2), (2, 3), (3, 3)\}$ be a relation on $A = \{1, 2, 3\}$, then R_2 is not a reflexive relation since $2 \in A$, $(2, 2) \notin R_2$.
 - (c) $R_3 = \{(x, y) \in \mathbb{R}^2 : x \leq y\}$ is a reflexive relation since $x \leq x$ for any $x \in \mathbb{R}$ (a set of real numbers).

Irreflexive Relation:

- A relation R on a set A is irreflexive if, for every $a \in A$, $(a, a) \notin R$. In other words, there is no $a \in A$ such that aRa .

Note:

The terms reflexive and irreflexive are extreme cases. Reflexive means that aRa is true for all a , and irreflexive means that aRa is true for no a .

- For ex.,

- (a) The relation $R_1 = \{(1,2), (1,3), (2,1), (2,3)\}$ on $A = \{1,2,3\}$ is irreflexive relation since $(x,x) \notin R_1$ for every $x \in A$.
- (b) The relation $R_2 = \{(x,y) \in \mathbb{R}^2 : x < y\}$ is an irreflexive relation since $x < x$ for no $x \in \mathbb{R}$.

Non-reflexive relation :

- A relation R on a set A is non-reflexive if R is neither reflexive nor irreflexive i.e., if aRa is true for some 'a' and false for others.
- For ex., $R = \{(1,2), (2,3), (2,2), (3,1)\}$ on $A = \{1,2,3\}$ is a non-reflexive relation since $2R2$ is true but $1R1$ and $3R3$ are false.

Symmetric Relation :

- A relation R on a set A is symmetric if whenever $(a,b) \in R$ then $(b,a) \in R$ i.e., if $aRb \Rightarrow bRa$.

- For ex.,

- (a) $R_1 = \{(1,1), (1,2), (1,3), (2,2), (2,1), (3,1)\}$ on $A = \{1,2,3\}$ is a symmetric relation.

- (b) $R_2 = \{(x,y) \in \mathbb{R}^2 : x^2 + y^2 = 1\}$ is a symmetric relation on $\mathbb{R}$ since if $x^2 + y^2 = 1$ then $y^2 + x^2 = 1$ too i.e., if $(x,y) \in R_2$ then $(y,x) \in R_2$.

Asymmetric Relation :

- A relation R on a set A is asymmetric if whenever $(a,b) \in R$ then $(b,a) \notin R$ for $a \neq b$, i.e., if $aRb \Rightarrow b \not R a$.
- For ex., $R = \{(1,1), (1,2), (2,3), (3,1)\}$ on $A = \{1,2,3\}$ is a asymmetric relation.

Antisymmetric Relation:

- A relation R on a set A is antisymmetric if for all $a, b \in A$
 $(aRb \text{ and } bRa) \Rightarrow a = b$. [In a formal way, relation R is antisymmetric, specifically if for all a and b in A , if $R(a, y)$ with $a \neq y$, then $R(y, x)$ must not hold, or equivalently, if $R(x, y)$ and $R(y, x)$, then $a = y$]
- For ex,

(a) $R_1 = \{(1, 2), (2, 2), (2, 3)\}$ on $A = \{1, 2, 3\}$ is an antisymmetric relation.

(b) $R_2 = \{(x, y) \in \mathbb{R}^2 : x \leq y\}$ is an antisymmetric relation on $\mathbb{R}$
 since $x \leq y$ and $y \leq x$ implies $x = y$, then $(x, y) \in R$
 and $(y, x) \in R$ implies $x = y$.

Note:

Antisymmetric is not the same as not symmetric.

A relation may be symmetric as well as antisymmetric at the same time. For ex, the relation $R = \{(1, 1), (3, 3)\}$ is both symmetric and antisymmetric on $A = \{1, 2, 3\}$.

Transitive Relation:

- A relation R on a set A is transitive if whenever
 $(a, b) \in R$ and $(b, c) \in R$, then $(a, c) \in R$, i.e., aRb and $bRc \Rightarrow aRc$.

- For ex,

(a) The relation 'is parallel to' on the set of lines in a plane is transitive, because if a line x is parallel to line y and if y is parallel to line z , then x is parallel to z .

(b) The relations 'is less than' and 'is greater than' are transitive relations on the set of real numbers. If $a < b$ and $b < c$ implies $a < c$ and if $a > b$ and $b > c$ implies $a > c$ for all real numbers a, b, c .

Let $A = \{1, 2, 3\}$. Let R be a relation defined on A .
Give an example of a relation which is

- (i) reflexive and transitive but not symmetric.
- (ii) symmetric and transitive but not reflexive.
- (iii) reflexive and symmetric but not transitive.

→

(i) $R_1 = \{(1,1), (2,2), (3,3), (1,3)\}$

is reflexive and transitive, but not symmetric, since $(1,3) \in R_1$ but $(3,1) \notin R_1$

(ii) $R_2 = \{(1,1), (3,3), (1,3), (3,1)\}$

is symmetric and transitive, but not reflexive, since $(2,2) \notin R_2$

(iii) $R_3 = \{(1,1), (2,2), (3,3), (1,2), (2,1), (2,3), (3,2)\}$

is reflexive and symmetric but not transitive since $(1,2) \in R_3$ and $(2,3) \in R_3$ but $(1,3) \notin R_3$.

Equivalence Relation:

- A relation R on a set A is called an equivalence relation if it is reflexive, symmetric and transitive.
- That is, R is an equivalence relation on A if it has the following three properties:

- (1) $(a,a) \in R$ for all $a \in A$ (reflexive)
- (2) $(a,b) \in R$ implies $(b,a) \in R$ (symmetric)
- (3) (a,b) and $(b,c) \in R$ imply $(a,c) \in R$ (transitive)

Ex: Let R is the relation on the set of strings of Hindi letters such that aRb if and only if $l(a) = l(b)$, where $l(x)$ is the length of the string x . Show that R is an equivalence relation.

→ Since $l(a) = l(a)$, it follows that aRa whenever ' a ' is a string, i.e. R is reflexive.

Suppose aRb , so that $l(a) = l(b)$. Then bRa , since $l(b) = l(a)$.
Hence, R is symmetric.

Again suppose that aRb and bRc , i.e., $l(a) = l(b)$ and $l(b) = l(c)$.
Hence $l(a) = l(c)$ which implies aRc . Consequently, R is transitive.

Since R is reflexive, symmetric and transitive, it is an equivalence relation.

Ex: If R be a relation in the set of integers $\mathbb{Z}$ defined by
 $R = \{(x, y) : x \in \mathbb{Z}, y \in \mathbb{Z}, (x-y) \text{ is divisible by } 6\}$
Then prove that R is an equivalence relation.

Let $x \in \mathbb{Z}$. Then $x - x = 0$ and 0 is divisible by 6 .

$\therefore xRx$ for all $x \in \mathbb{Z}$.

Hence, R is reflexive.

Again, $xRy \Rightarrow (x-y)$ is divisible by 6

$\Rightarrow -(x-y)$ is divisible by 6

$\Rightarrow (y-x)$ is divisible by 6

$\Rightarrow yRx$

Hence, R is symmetric.

xRy and $yRz \Rightarrow (x-y)$ is divisible by 6 and $(y-z)$ is divisible

$\Rightarrow [(x-y) + (y-z)]$ is divisible by 6

$\Rightarrow (x-z)$ is divisible by 6

$\Rightarrow xRz$

Hence, R is transitive.

Thus, R is an equivalence relation. (proved).

Ex: Consider the following relation on $\{1, 2, 3, 4, 5, 6\}$

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$$R = \{(i, j) : |i - j| = 2\}$$

Is 'R' transitive? Is R reflexive? Is R symmetric?

→ Let $A = \{1, 2, 3, 4, 5, 6\}$

Then $R = \{(i, j) : |i - j| = 2\}$ on A

$$= \{(1, 3), (2, 4), (3, 1), (4, 2), (3, 5), (5, 3), (4, 6), (6, 4)\}$$

R is not reflexive since $(i, i) \notin R \forall i \in A$. For ex., $(1, 1) \notin R$.

R is symmetric since for all $(i, j) \in R$, (j, i) also $\in R$.

R is not transitive since for all (i, j) and $(j, k) \in R$, (i, k) does not belong to R. For ex., $(2, 4)$ and $(4, 2) \in R$ but $(2, 2) \notin R$.

Ex: Let R be a binary relation on the set of all integers of 0's and 1's such that $R = \{(a, b) : a \text{ and } b \text{ are strings, that they have the same number of 0's}\}$. Is R reflexive? symmetric? Transitive? An equivalence relation?

→ R is reflexive since $(a, a) \in R \forall a \in R$.

R is symmetric since when a and b have the same number of 0's then b and a will also have the same number of 0's. Hence $(a, b) \in R \Rightarrow (b, a) \in R$.

R is transitive since when a and b have the same number of 0's and b and c have the same number of 0's, then a and c will also have the same number of 0's. Hence (a, b) and $(b, c) \in R \Rightarrow (a, c) \in R$.

Thus, R is reflexive, symmetric and transitive and hence an equivalence relation.

Theorem: Let R and S be relation from A to B , show that

(i) If $R \subseteq S$, then $R^{-1} \subseteq S^{-1}$.

(ii) $(R \cap S)^{-1} = R^{-1} \cap S^{-1}$.

(iii) $(R \cup S)^{-1} = R^{-1} \cup S^{-1}$.

Proof:

i) Suppose $R \subseteq S$. If $(a, b) \in R^{-1}$, then $(b, a) \in R$ and also $(b, a) \in S$ since $R \subseteq S$. Again $(b, a) \in S$ implies $(a, b) \in S^{-1}$. Therefore $R^{-1} \subseteq S^{-1}$. (proved)

ii) Let $(a, b) \in (R \cap S)^{-1}$. Then $(b, a) \in R \cap S$, so that $(b, a) \in R$ and $(b, a) \in S$. This implies $(a, b) \in R^{-1}$ and $(a, b) \in S^{-1}$. Hence $(a, b) \in R^{-1} \cap S^{-1}$. Therefore, $(R \cap S)^{-1} \subseteq R^{-1} \cap S^{-1}$ — (1)

Conversely, let $(a, b) \in R^{-1} \cap S^{-1}$. Then $(a, b) \in R^{-1}$ and $(a, b) \in S^{-1}$. This implies $(b, a) \in R$ and $(b, a) \in S$. So $(b, a) \in R \cap S$. Hence $(a, b) \in (R \cap S)^{-1}$. Therefore, $R^{-1} \cap S^{-1} \subseteq (R \cap S)^{-1}$ — (2)

From (1) and (2), we have,

$$(R \cap S)^{-1} = R^{-1} \cap S^{-1} \text{ (proved)}$$

iii) Let $(a, b) \in (R \cup S)^{-1}$. Then $(b, a) \in R \cup S$ so that $(b, a) \in R$ or $(b, a) \in S$. This implies $(a, b) \in R^{-1}$ or $(a, b) \in S^{-1}$. Hence, $(a, b) \in R^{-1} \cup S^{-1}$. Therefore, $(R \cup S)^{-1} \subseteq R^{-1} \cup S^{-1}$ — (1)

Conversely, let $(a, b) \in R^{-1} \cup S^{-1}$. Then $(a, b) \in R^{-1}$ or $(a, b) \in S^{-1}$. This implies $(b, a) \in R$ or $(b, a) \in S$. So, $(b, a) \in R \cup S$. Hence, $(a, b) \in (R \cup S)^{-1}$. Therefore, $R^{-1} \cup S^{-1} \subseteq (R \cup S)^{-1}$ — (2)

From (1) and (2), we get,

$$(R \cup S)^{-1} = R^{-1} \cup S^{-1} \text{ (proved)}$$

Theorem: Let R be a relation on A . Prove that

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(i) If R is reflexive, so is R^{-1} .

(ii) R is symmetric if and only if $R = R^{-1}$.

~~(iii) R is transitive if and only if $R \circ R \subseteq R$.~~

Proof:

(i) Suppose R is reflexive. Then $(a, a) \in R$ for all $a \in A$.

So, $(a, a) \in R^{-1}$ for all $a \in A$. Therefore, R^{-1} is reflexive. (proved)

(ii) Suppose R is symmetric. Let $(a, b) \in R^{-1}$. Then $(b, a) \in R$ and hence $(a, b) \in R$ since R is symmetric. Therefore $R^{-1} \subseteq R$. Similarly, it can be shown that $R \subseteq R^{-1}$. Hence $R = R^{-1}$.

Conversely, suppose $R = R^{-1}$. Let $(a, b) \in R$. Then $(a, b) \in R^{-1}$ and so $(b, a) \in R$. Hence, R is symmetric. Thus, R is symmetric if and only if $R = R^{-1}$. (proved).

Theorem: Suppose R and S are relations on a set A . Prove that

(i) If R and S are reflexive, then $R \cup S$ and $R \cap S$ are reflexive.

(ii) If R and S are symmetric, then $R \cup S$ and $R \cap S$ are symmetric.

(iii) If R and S are transitive, then $R \cap S$ is transitive.

Proof:

(i) Suppose R and S are reflexive. Then $(a, a) \in R$ and $(a, a) \in S$ for all $a \in A$. Therefore, $(a, a) \in R \cup S$ and $(a, a) \in R \cap S$.

Hence, $R \cup S$ and $R \cap S$ are reflexive. (proved)

(ii) Suppose R and S are symmetric. Then, we know, $R = R^{-1}$ and $S = S^{-1}$. Using $(R \cup S)^{-1} = R^{-1} \cup S^{-1}$, we can write $(R \cup S)^{-1} = R \cup S$ and using $(R \cap S)^{-1} = R^{-1} \cap S^{-1}$, we can write $(R \cap S)^{-1} = R \cap S$. Hence, $R \cup S$ and $R \cap S$ are symmetric. (proved)

(2) transitive. (a,b) (b,c) $\in R^{-1} \Rightarrow (a,c) \in R^{-1}$

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(iii) Suppose R and S are transitive. Let $(a,b) \in R \cap S$ and $(b,c) \in R \cap S$, then $(a,b) \in R$, $(a,b) \in S$, $(b,c) \in R$ and $(b,c) \in S$.
Now, $(a,b) \in R$ and $(b,c) \in R$ implies $(a,c) \in R$ and $(a,b) \in S$ and $(b,c) \in S$ implies $(a,c) \in S$. Therefore, $(a,c) \in R \cap S$.
Hence, $(a,b) \in R \cap S$ and $(b,c) \in R \cap S$ implies $(a,c) \in R \cap S$.
Therefore, $R \cap S$ is transitive. (proved)

Note:

If R and S are transitive relations on A , then $R \cup S$ need not be a transitive relation on A . For ex,
if $R = \{(1,1), (2,2), (1,2), (2,1)\}$ and $S = \{(2,2), (3,3), (2,3), (3,2)\}$ on $A = \{1,2,3\}$, then R and S are transitive but $R \cup S$ is not transitive since $(1,2) \in R \cup S$ and $(2,3) \in R \cup S$ but $(1,3) \notin R \cup S$.
Hence, $R \cup S$ is not transitive.

Theorem: If R and S are equivalence relations on the set A , prove that

- (i) R^{-1} is an equivalence relation.
- (ii) $R \cap S$ is an equivalence relation.

Proof:

(i) Let R be an equivalence relation on a set A .
Therefore, R is reflexive, symmetric and transitive.
Let $a, b, c \in A$ be arbitrary.

The relation R^{-1} is

(1) reflexive: $(a,a) \in R^{-1}$, since $(a,a) \in R$ for all $a \Rightarrow (a,a) \in R^{-1}$

(2) symmetric: $(a,b) \in R^{-1} \Rightarrow (b,a) \in R^{-1}$

Since, $(a,b) \in R^{-1} \Rightarrow (b,a) \in R$

$\Rightarrow (a,b) \in R$, as R is symmetric
 $\Rightarrow (b,a) \in R^{-1}$.



(3) transitive : $(a, b), (b, c) \in R^{-1} \Rightarrow (a, c) \in R^{-1}$ (31)

Since, $(a, b), (b, c) \in R^{-1} \Rightarrow (b, a), (c, b) \in R$
 $\Rightarrow (c, b), (b, a) \in R$
 $\Rightarrow (c, a) \in R$, as R is transitive
 $\Rightarrow (a, c) \in R^{-1}$.

$\therefore R^{-1}$ is reflexive, symmetric and transitive.

Hence, R^{-1} is an equivalence relation on A . (proved)

ii) We have to verify that $R \cap S$ is reflexive, symmetric and transitive.

(1) For all $a \in A$, $(a, a) \in R$ and $(a, a) \in S$, since R and S are equivalence relation. Hence for all $a \in A$, $(a, a) \in R \cap S$.
Hence, $R \cap S$ is reflexive.

(2) $(a, b) \in R \cap S \Rightarrow (a, b) \in R$ and $(a, b) \in S$
 $\Rightarrow (b, a) \in R$ and $(b, a) \in S$, since R and S are symmetric being equivalence relations.

$\Rightarrow (b, a) \in R \cap S$.

Hence, $R \cap S$ is symmetric.

(3) $(a, b) \in R \cap S$, $(b, c) \in R \cap S \Rightarrow (a, b) \in R$, $(b, c) \in R$ and $(a, b) \in S$, $(b, c) \in S$
 $\Rightarrow (a, c) \in R$ and $(a, c) \in S$, $\therefore$ transitive since R and S are transitive being equivalence relations.

$\Rightarrow (a, c) \in R \cap S$.

Hence, $R \cap S$ is transitive.

$\therefore R \cap S$ is an equivalence relation on A . (proved)



Equivalence classes:

- If R is an equivalence relation on a set S and xRy , then x and y are called equivalent with respect to R .
- The set of all elements of S that are equivalent to a given element x constitute the equivalence class of x , denoted by $[x]_R$.
- When only one relation is under consideration, the subscript R is deleted and the equivalence class is denoted by $[x]$.

$$\text{Thus, } [x] = \{y \in S : yRx\}$$

- For ex., the equivalence relation

$$R = \{(1,2), (2,1), (1,1), (2,2), (3,3), (4,4)\} \text{ on } S = \{1,2,3,4\}$$

has the following equivalence classes

$$[1] = [2] = \{1,2\}$$

$$[3] = \{3\}$$

$$[4] = \{4\}$$

Important properties of equivalence classes:

Let A be a non-empty set and R be an equivalence relation defined in A . Let a and $b \in A$ be arbitrary.

Then

$$(i) \ a \in [a]$$

$$(ii) \ b \in [a] \Rightarrow [b] = [a]$$

$$(iii) \ [a] = [b] \Leftrightarrow (a,b) \in R$$

Ex: Show that the relation $(x, y) R (a, b) \Leftrightarrow x^2 + y^2 = a^2 + b^2$ (33)
is an equivalence relation on the plane and describe the
equivalence classes.

→
$$\begin{aligned}(x, y) R (a, b) &\Leftrightarrow x^2 + y^2 = a^2 + b^2 \\ &\Leftrightarrow a^2 + b^2 = x^2 + y^2 \\ &\Leftrightarrow (a, b) R (x, y)\end{aligned}$$

Hence, R is symmetric.

Now, $(x, y) R (a, b)$ and $(a, b) R (c, d)$
$$\begin{aligned}&\Leftrightarrow x^2 + y^2 = a^2 + b^2 \text{ and } a^2 + b^2 = c^2 + d^2 \\ &\Leftrightarrow x^2 + y^2 = c^2 + d^2 \\ &\Leftrightarrow (x, y) R (c, d)\end{aligned}$$

Hence, R is transitive.

Again, $(x, y) R (x, y) \Leftrightarrow x^2 + y^2 = x^2 + y^2$

Hence, R is reflexive.

Thus, R is symmetric, transitive and reflexive and hence
an equivalence relation.

2nd part:

Now, for any point (x, y) the sum $x^2 + y^2$ is the
square of its distance from the origin. The equivalence
classes are, therefore, the sets of points in the plane
which have the same distance from the origin. Thus,
the equivalence classes are concentric circles centred on
the origin.

Composition of Relations

Let A, B, C be sets. Let R be a relation from A to B and let S be a relation from B to C . That is R is a subset of $A \times B$ and S is a subset of $B \times C$. The composite of R and S , denoted by $R \circ S$, is the relation consisting of ordered pairs (a, c) where $a \in A$ and $c \in C$, and for which there exists an element $b \in B$ such that $(a, b) \in R$ and $(b, c) \in S$.

Thus, $R \circ S = \{(a, c) \in A \times C : \text{for some } b \in B, (a, b) \in R \text{ and } (b, c) \in S\}$

That is $a(R \circ S) \subseteq$ if for some $b \in B$ we have $a R b$ and $b S c$.

Note: (1) $R \circ S$ is empty if the intersection of the range of R and the domain of S is empty. For the relation $R \circ S$, the domain is a subset of A and the range is a subset of C .

(2) If R be a relation from A to B , S is a relation from B to C , P be a relation from C to D , then $R \circ (S \circ P)$ and $(R \circ S) \circ P$ are relations from A to D . The operation of composition on relations is associative i.e., $R \circ (S \circ P) = (R \circ S) \circ P$.

Ex: Let $(a, b) \in R$, $(b, c) \in S$ and $(c, d) \in P$.

Then $(b, d) \in S \circ P$ and $(a, d) \in R \circ (S \circ P)$

Again $(a, c) \in R \circ S$ and $(a, d) \in (R \circ S) \circ P$

This shows $R \circ (S \circ P) = (R \circ S) \circ P$.

(3) The composition of a relation with itself is denoted with a power of a relation R . Let R be a relation on the set A . Then $R \circ R$ the composition of R with itself and $R \circ R$ is denoted by R^2 . Similarly, $R^3 = R \circ R \circ R$ and so on.

For any relation R and natural number i one defines R^i as

$R^0 = I$, the identity relation

$R^i = R \circ R^{i-1}$, for $i > 0$.

Ex: Let $A = \{1, 2, 3\}$, $B = \{p, q, r\}$, $C = \{x, y, z\}$ and let
 $R = \{(1, p), (1, r), (2, q), (3, q)\}$ and $S = \{(p, x), (q, x), (r, z)\}$.

Compute $R \circ S$.

→ $R \circ S = \{(1, x), (1, z), (2, x), (3, x)\}$

Ex: Let $R = \{(1, 1), (2, 1), (3, 2)\}$, compute R^2

→ $R^2 = R \circ R$
 $= \{(1, 1), (2, 1), (3, 2)\} \circ \{(1, 1), (2, 1), (3, 2)\}$
 $= \{(1, 1), (2, 1), (3, 1)\}$

Theorem: If R_1 and R_2 are relations from A to B and R_3 and R_4 are relations from B to C , then

i) If $R_1 \subseteq R_2$ and $R_3 \subseteq R_4$ then $R_1 \circ R_3 \subseteq R_2 \circ R_4$

ii) $(R_1 \cup R_2) \circ R_3 = R_1 \circ R_3 \cup R_2 \circ R_3$.

i) Let $(x, z) \in R_1 \circ R_3$

Then for some $y \in B$, we have, $(x, y) \in R_1$ and $(y, z) \in R_3$

Since, $R_1 \subseteq R_2$ and $R_3 \subseteq R_4$, we also have

$(x, y) \in R_2$ and $(y, z) \in R_4$.

So, $(x, z) \in R_2 \circ R_4$

This shows that $R_1 \circ R_3 \subseteq R_2 \circ R_4$. (proved)

ii) We have, $R_1 \circ R_3 \subseteq (R_1 \cup R_2) \circ R_3$ [from i)]

Similarly, $R_2 \circ R_3 \subseteq (R_1 \cup R_2) \circ R_3$

and so, $R_1 \circ R_3 \cup R_2 \circ R_3 \subseteq (R_1 \cup R_2) \circ R_3$

Consider, $(x, z) \in (R_1 \cup R_2) \circ R_3$. Then for some $y \in B$, we have,

$(x, y) \in R_1 \cup R_2$ and $(y, z) \in R_3$

$(x, y) \in R_2$, so that.

Then either $(x, y) \in R_1$ so that $(x, z) \in R_1 \circ R_3$

or ~~else~~ $(x, z) \in R_2 \circ R_3$.

Either way $(x, z) \in R_1 \circ R_3 \cup R_2 \circ R_3$.

Theorem: If R is a relation from A to B and S is a relation from B to C , show that $(R \circ S)^{-1} = S^{-1} \circ R^{-1}$. (31)

Proof: Since R is a relation from A to B , R^{-1} is a relation from B to A . Similarly, S^{-1} is a relation from C to B . Also the relation $S^{-1} \circ R^{-1}$ is from C to A . If xRy and ySz , then $x(R \circ S)z$ and $x(R \circ S)^{-1}z$. But $zS^{-1}y$ and $yR^{-1}x$ so that $z(S^{-1} \circ R^{-1})x$. This is true for any $x \in A$ and $z \in C$; hence, the required result.

Theorem: If R is a relation on a set A , then R is transitive if and only if $R^2 \subseteq R$.

Proof: Let R is transitive and $(x, z) \in R^2$. By definition of R^2 there exists $y \in A$ such that $(x, y) \in R$ and $(y, z) \in R$. Since R is transitive (x, z) is also in R . That is every $(x, z) \in R^2$ is in R .

Therefore, $R^2 \subseteq R$.

Conversely, suppose $R^2 \subseteq R$. Let (x, y) and $(y, z) \in R$. Then $(x, z) \in R^2$ and hence in R . This proves that R is transitive.

Matrix Representation of Composition:

The matrix for the composite of ~~matrix~~ relations can be found using the Boolean product of the matrices. Suppose, R is a relation from A to B and S be a relation from B to C . Suppose that A, B and C have elements m, n and p respectively. The matrices represented by R, S and $R \circ S$ are denoted by $M_R = [r_{ij}]_{m \times n}$, $M_S = [s_{ij}]_{n \times p}$ and $M_{R \circ S} = [t_{ij}]_{m \times p}$ respectively. It follows that $t_{ij} = 1$ if and only if $r_{ik} = s_{kj} = 1$ for some k . From the definition of Boolean product, this means that $M_{R \circ S} = M_R \cdot M_S$.

note:

i) Since the Boolean matrix multiplication is not commutative ⁽³²⁾ in general so the composition of relations is not commutative in general.
i.e., $R \circ S \neq S \circ R$.

ii) The Boolean matrix ^{multiplication} is associative so composition on relations is also associative, that is

$$R \circ (S \circ T) = (R \circ S) \circ T$$

iii) The distributive law is valid for Boolean matrix multiplication so composition of relation also satisfies the distributive law, that is

$$(R \cup S) \circ T = R \circ T \cup S \circ T$$

Ex:

$$M_R = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad M_S = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$M_{R \circ S} = M_R \cdot M_S = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \cdot \begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Closure of Relations:

Let R be a relation on a set A . R may or may not have some property P , such as reflexivity, symmetry or transitivity. If there is a relation S with property P containing R such that S is a subset of every relation with P containing R , then S is called the closure of R with respect to P .

Reflexive closure:

The reflexive closure $R(r)$ of a relation R can be formed by adding to R all pairs of the form (a, a) with $a \in R$, not already in R (since for reflexive closure xRx to be true for all x). Thus

$$R(r) = R \cup I_A$$

Ex: The reflexive closure of $<$ (less than) relation on ~~real~~ real (33) numbers is obtained by adding the identity relation which is the equality relation on real numbers and the union of $<$ and $=$ is $\leq$. Thus, the ~~relation~~ $\leq$ relation is the reflexive closure of the relation $<$.

Symmetric closure:

A symmetric relation contains (x, y) if it contains (y, x) . Since the inverse relation R^{-1} contains (y, x) if (x, y) is in R , the symmetric closure $R^{(s)}$ of a relation R can be constructed by taking the union of R and R^{-1} , that is $R \cup R^{-1}$ is the symmetric closure of R , ~~that is~~

$$\text{i.e., } R^{(s)} = R \cup R^{-1}$$

Ex: If $R = \{(1, 2), (4, 3), (2, 2), (2, 1), (3, 1)\}$ be a relation on $S = \{1, 2, 3, 4\}$, find the symmetric closure.

$$\rightarrow R^{-1} = \{(2, 1), (3, 4), (2, 2), (1, 2), (1, 3)\}$$

$$\text{So, } R^{(s)} = R \cup R^{-1} = \{(1, 2), (2, 1), (4, 3), (3, 4), (3, 1), (1, 3), (2, 2)\}$$

Transitive closure: The relation obtained by adding the least number of ordered pairs to ensure transitivity is called the transitive closure of the relation. The transitive closure of R is denoted by R^+ . Let a relation R is defined on A and A contains n elements. Consequently, to make a relation R transitive, one has to add all pairs of R^2 , all pairs of R^3 , ..., all pairs of R^n , unless these pairs are already in R . Thus R^+ is calculated as the union of the terms of the form R^k .

$$R^+ = R \cup R^2 \cup \dots \cup R^n.$$

In matrix terminology, if M denotes the relational matrix of a relation then the symmetric closure of R , denoted by M_S , can be obtained from

$$M_S = M \vee M^T.$$

The reflexive closure of R , denoted by M_R , can be obtained from

$$M_R = M \vee I_n.$$

The transitive closure of R , denoted by M_T , can be obtained from

$$M_T = M \vee M^2 \vee M^3 \vee \dots \vee M^n.$$

The closure of relation R in graphical form is given as follows:

Reflexive Closure: We add all the missing arrows from points to themselves in the digraph of R .

Symmetric Closure: We add missing ~~arrows~~ ^{reverses} of all the arrows in the digraph of R .

Transitive Closure: We add an arrow connecting a point x to y whenever some sequence of arrows in the digraph of R connected x to y and there was not an arrow from x to y already.

Ex: Let $R = \{(1, 2), (2, 3), (3, 1)\}$ and $A = \{1, 2, 3\}$, find ⁽³⁾ the reflexive, symmetric and transitive closure of R , using

- i) Composition of relation R
- ii) Composition of matrix relation R
- iii) Graphical representation of R .

i) ~~The reflexive closure~~

$$R^{(1)} = R \cup I_A = \{(1, 2), (2, 3), (3, 1)\} \cup \{(1, 1), (2, 2), (3, 3)\} \\ = \{(1, 1), (1, 2), (2, 2), (2, 3), (3, 1), (3, 3)\}$$

$$R^{(2)} = R \cup R^{-1} = \{(1, 2), (2, 3), (3, 1)\} \cup \{(2, 1), (3, 2), (1, 3)\} \\ = \{(1, 2), (1, 3), (2, 1), (2, 3), (3, 1), (3, 2)\}$$

$$\text{Now, } R \circ R = \{(1, 2), (2, 3), (3, 1)\} \circ \{(1, 2), (2, 3), (3, 1)\}$$

$$R^2 = \{(1, 3), (2, 1), (3, 2)\}$$

$$R^3 = R^2 \circ R = \{(1, 3), (2, 1), (3, 2)\} \circ \{(1, 2), (2, 3), (3, 1)\} \\ = \{(1, 1), (2, 2), (3, 3)\}$$

$$R^4 = R^3 \circ R = \{(1, 1), (2, 2), (3, 3)\} \circ \{(1, 2), (2, 3), (3, 1)\} \\ = \{(1, 2), (2, 3), (3, 1)\} = R$$

$$\text{Thus } R^5 = R^4 \circ R = R \circ R = R^2$$

$$R^6 = R^5 \circ R = R^2 \circ R = R^3 \text{ and so on.}$$

$$\text{Hence, } R^+ = R \cup R^2 \cup R^3 = \{(1, 1), (1, 2), (1, 3), (2, 1), (2, 2), (2, 3), \\ (3, 1), (3, 2), (3, 3)\}$$

ii) Let M be the relation matrix of R . Then

36

$$M = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$$

$$M_S = M \vee M^T = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} \vee \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$$

$$R^{(S)} = \{(1,2), (1,3), (2,1), (2,3), (3,1), (3,2)\}$$

$$M_R = M \vee I_3 = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} \vee \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}$$

$$R^{(R)} = \{(1,1), (1,2), (2,2), (2,3), (3,1), (3,3)\}$$

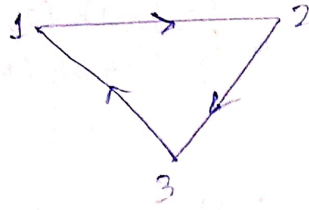
Now, $M^2 = M \circ M = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} \circ \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$

$$M^3 = M^2 \circ M = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \circ \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

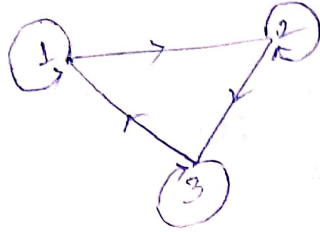
$$M_T = M \vee M^2 \vee M^3 = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} \vee \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \vee \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

$$R^+ = \{(1,1), (1,2), (1,3), (2,1), (2,2), (2,3), (3,1), (3,2), (3,3)\}$$

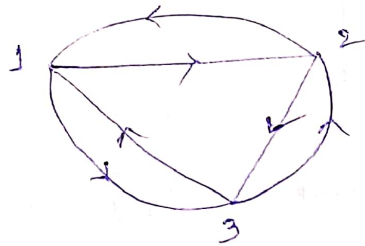
7) The graphical representation of R is shown in the figure below (37)



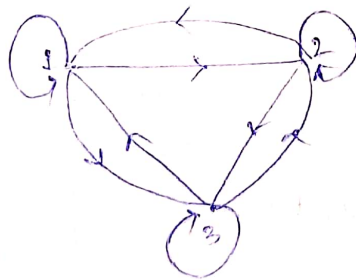
To find out the reflexive closure representation of R we add all the arrows from points to themselves. This is shown in the figure below



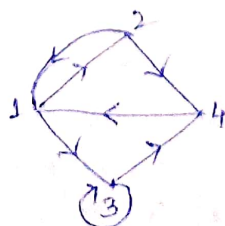
To find out the symmetric closure representation of R , we add reverse of all the arrows in graphical representation of R . This is shown in the figure below.



To find transitive arrow, we add arrow 1 to 1 since $1 \rightarrow 2 \rightarrow 3 \rightarrow 1$. Similarly, 2 to 2 and 3 to 3. Again we add arrow 1 to 3, since $1 \rightarrow 2 \rightarrow 3$. Similarly, 2 to 1 and 3 to 2. This is shown in the figure below.



Ex: Consider a relation R whose directed graph is shown in the figure below. Determine its inverse R^{-1} and complement R' .



$$R = \{(1, 2), (1, 3), (2, 1), (2, 4), (3, 4), (3, 2), (4, 1)\}$$

→ The directed graph of R^{-1} can be obtained from directed graph of R by reversing each arc.

(39)

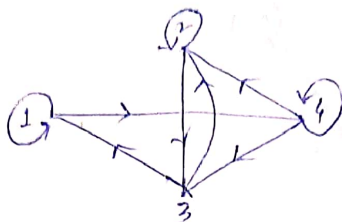
The universal relation $R \times R$ on the set $\{1, 2, 3, 4\}$ is

$$R \times R = \{(1, 1), (1, 2), (1, 3), (1, 4), (2, 1), (2, 2), (2, 3), (2, 4), (3, 1), (3, 2), (3, 3), (3, 4), (4, 1), (4, 2), (4, 3), (4, 4)\}$$

Thus, the complement of R is

$$R' = \{(1, 1), (1, 4), (2, 2), (2, 3), (3, 2), (3, 1), (4, 4), (4, 3), (4, 2)\}$$

The directed graph of R' is shown in the figure below



Ex: Consider a relation R defined on $A = \{1, 2, 3\}$ whose matrix representation is given below. Determine its inverse R^{-1} and complement R' .

$$M_R = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

→ We have,

$$M_{R^{-1}} = M_R^T \text{ (i.e., transpose of } M_R)$$

Now, $R = \{(1, 1), (2, 1), (2, 2), (2, 3), (3, 3)\}$

$$M_R^T = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$$

$$\therefore R^{-1} = \{(1, 1), (1, 2), (2, 2), (3, 2), (3, 3)\}$$

Again, we know $M_{R'}$ can be obtained by changing 0 into 1 and 1 into 0 in M_R .

$$M_{R'} = \begin{bmatrix} 0 & 1 & 1 \\ 0 & 0 & 0 \\ 1 & 1 & 0 \end{bmatrix}$$

$$\therefore R' = \{(1, 2), (1, 3), (3, 1), (3, 2)\}$$

Ex: Let $A = \{1, 2, 3\}$ and $B = \{a, b, c, d\}$. Let R and S be the relations from A to B with the Boolean matrices.

$$M_R = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 \end{bmatrix} \quad \text{and} \quad M_S = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix}$$

- Find Boolean matrices for R^{-1} and S^{-1} .
- Find Boolean matrices for $(R \cap S) \circ R^{-1}$ and $R \circ R^{-1} \cap S \circ R^{-1}$.
- Find Boolean matrices for $S \circ (R^{-1} \cup S^{-1})$ and $S \circ R^{-1} \cup S \circ S^{-1}$.
- Using (b) and (c) show that $(R \cap S) \circ R^{-1} \neq R \circ R^{-1} \cap S \circ R^{-1}$ and $S \circ (R^{-1} \cup S^{-1}) = S \circ R^{-1} \cup S \circ S^{-1}$.

→

$$a) \quad M_{R^{-1}} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad , \quad M_{S^{-1}} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

$$b) \quad M_{R \cap S} = M_R \wedge M_S = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 \end{bmatrix} \wedge \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$M_{(R \cap S) \circ R^{-1}} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$M_{R \circ R^{-1}} = M_R \cdot M_{R^{-1}} = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

$$M_{S \circ R^{-1}} = M_S \cdot M_{R^{-1}} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$$

$$\therefore M_{R \circ R^{-1}} \wedge M_{S \circ R^{-1}} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \wedge \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

The

$$\textcircled{b)} \Rightarrow M_{R^{-1}US^{-1}} = M_R^{-1} \vee M_S^{-1} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \vee \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$$

Thus

$$M_{S_0(R^{-1}US^{-1})} = M_S \cdot M_{R^{-1}US^{-1}} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix} \cdot \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

The

Again, $M_{S_0R^{-1}} = M_S \cdot M_R^{-1} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$

Consi

$$M_{S_0S^{-1}} = M_S \cdot M_{S^{-1}} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix} \cdot \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

is

$$M_{S_0R^{-1}} \vee M_{S_0S^{-1}} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix} \vee \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

We

$\Rightarrow$ From (b), we see

$$M_{(R \cap S)_0R^{-1}} \neq M_{R_0R^{-1}} \wedge M_{S_0S^{-1}}$$

$$\therefore (R \cap S)_0R^{-1} \neq R_0R^{-1} \cap S_0R^{-1}$$

And from (c), we see

$$M_{S_0(R^{-1}US^{-1})} = M_{S_0R^{-1}} \vee M_{S_0S^{-1}}$$

$$\therefore S_0(R^{-1}US^{-1}) = S_0R^{-1} \vee S_0S^{-1}$$

```

1. printf("%d", top);
2. for (i=0; i <= top; i++)
3. { printf("%d", stack[i]);
4. }
5. printf("\n\n");
6. getch();

```

3

Let $A = \{1, 2, 3\}$ and $B = \{1, 2, 3, 4\}$. Let R and S be two relations from A to B defined by $R = \{(1, 1), (2, 2), (3, 3)\}$ and $S = \{(1, 1), (1, 2), (1, 3), (1, 4)\}$. Find $R \cup S$, $R \cap S$, R^{-1} and S^{-1} and show that $(R \cap S)^{-1} = R^{-1} \cap S^{-1}$.

→ i) $\{(1, 1), (1, 2), (1, 3), (1, 4), (2, 2), (3, 3)\}$ ii) $\{(1, 1)\}$
 iii) $\{(1, 1), (2, 2), (3, 3)\}$ iv) $\{(1, 1), (2, 1), (3, 1), (4, 1)\}$

Q. The relation R on the set $\{1, 2, 3, 4, 5\}$ is defined by the rule $(x, y) \in R$ if x divides y . Find
 i) the elements of R ii) the elements of R^{-1} iii) the domain of R
 iv) the range of R v) the domain of R^{-1} vi) the range of R^{-1} .

→ i) $R = \{(1, 1), (1, 2), (1, 3), (1, 4), (1, 5), (2, 2), (2, 4), (2, 5), (3, 3), (4, 4), (5, 5)\}$
 ii) $R^{-1} = \{(1, 1), (2, 1), (3, 1), (4, 1), (5, 1), (2, 2), (3, 3), (4, 4), (5, 5), (4, 2), (5, 2), (5, 3)\}$
 iii) $\text{dom}(R) = \{1, 2, 3, 4, 5\}$
 iv) $\text{ran}(R) = \{1, 2, 3, 4, 5\}$
 v) $\text{dom}(R^{-1}) = \text{dom}(R)$
 vi) $\text{ran}(R^{-1}) = \text{ran}(R)$.

Q. The relation R on the set $\{1, 2, 3, 4, 5\}$ is defined by the rule $(x, y) \in R$ if $x + y \leq 6$. Find R and R^{-1} .

→ $R = R^{-1} = \{(1, 1), (1, 2), (1, 3), (1, 4), (1, 5), (2, 1), (2, 2), (2, 3), (2, 4), (3, 1), (3, 2), (3, 3), (4, 1), (4, 2), (5, 1)\}$

$\text{dom } R = \text{ran } R = \text{dom } R^{-1} = \text{ran } R^{-1} = \{1, 2, 3, 4, 5\}$.